

Yingrui Hou
Clermont-Ferrand, France
+33 (0)7 85 39 63 32
yingrui.hou@outlook.com
<https://yingrui-hou.github.io/PersonalPage/>

27 August 2026

Dr Francesco Pinotti and Dr Guillaume Fournié
UMR Épidémiologie des maladies animales et zoonotiques
INRAE / VetAgro Sup, Marcy-l'Étoile, France

Subject: Application for Postdoctoral Position OT-30377 - Epidemiological Modelling, Bayesian Inference and AI

Dear Dr Pinotti and Dr Fournié,

I am applying for postdoctoral position OT-30377 on epidemiological modelling, simulation-based inference, and artificial intelligence. I hold a PhD in experimental physics and have several years of experience developing Python/C++ workflows for probabilistic modelling, simulation calibration, latent-parameter inference, and uncertainty-aware validation. Extending these methods to mechanistic pathogen-transmission models closely matches the methodological problem solving that now motivates my research.

My work has repeatedly addressed inference when the system state is only indirectly observable. I implemented probabilistic and maximum-likelihood models under background contamination, finite resolution, selection effects, and dataset mismatch. I built calibration workflows integrating simulated, control, and measured datasets, and contributed covariance-aware checks, pseudoexperiments, and uncertainty-stress studies. Time-dependent analyses and monitoring workflows converted expected-versus-observed discrepancies into model, calibration, or software follow-up actions.

This experience provides a strong foundation for calibrating mechanistic models from heterogeneous observations and evaluating predictive uncertainty. I have not yet applied NPE or NLE to epidemiological models or used PyTorch, JAX, Pyro, sbi, or Stan professionally. I would address these focused gaps through implementation on epidemic simulators, simulation-based calibration, coverage checks, and comparison with likelihood-based baselines.

I am accustomed to autonomous research in Linux-based shared codebases using Python, C++, Git, reproducible workflows, documentation, and collaborative review. My postdoctoral work involved international collaborations, peer-reviewed publications, methodological notes, and scientific presentations, requiring clear communication across technical specialities.

My interest in moving beyond particle physics is deliberate: I value building reliable methods for difficult real-world inference problems. Applying this experience to epidemic preparedness and animal and human health would give that focus direct societal relevance. I would welcome the opportunity to discuss how my simulation calibration, latent-variable inference, and robust validation experience could contribute to GeoAI4EI and CRISPOX.

Yours sincerely,

Yingrui Hou